

BUDAPEST FERIHEGY, Hungary

WMO: 128390

Lat: 47.437N

Lon: 19.256E

Elev: 151

StdP: 99.53

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-9.0	-14.1	1.1	-9.5	-11.2	1.5	-7.6	12.9	6.6	11.3	5.1	1.5	320	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	33.1	22.0	31.2	21.1	29.6	20.5	23.2	30.5	22.2	29.0	21.2	27.6	4.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.9	15.9	27.3	20.0	15.0	25.5	19.1	14.1	24.2	69.7	30.5	65.6	28.9	62.3	27.5	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.0	7.7	DB	-15.6	35.5	3.2	2.0	-17.9	36.9	-19.8	38.0	-21.7	39.1	-24.0	40.6	(4)
(5)			WB	-15.9	24.6	3.2	1.5	-18.2	25.6	-20.0	26.5	-21.8	27.3	-24.1	28.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-0.5	1.3	5.7	11.3	16.0	19.6	21.6	21.2	16.2	10.9	5.6	0.2	(6)	
	DBStd	8.80	4.32	4.85	4.22	3.84	3.57	3.48	3.16	3.38	3.50	4.09	4.16	4.45	(7)	
	HDD10.0	1226	325	244	142	29	2	0	0	0	1	39	141	303	(8)	
	HDD18.3	3052	583	476	393	211	88	26	8	12	82	231	382	561	(9)	
	CDD10.0	1527	0	1	8	69	189	289	359	348	188	68	8	0	(10)	
	CDD18.3	311	0	0	0	1	17	64	108	101	19	1	0	0	(11)	
	CDH23.3	3443	0	0	0	29	236	722	1175	1075	197	9	0	0	(12)	
	CDH26.7	1231	0	0	0	3	46	247	463	426	46	0	0	0	(13)	
(14) Wind	WSAvg	3.1	3.0	3.2	3.6	3.6	3.3	3.2	3.3	2.8	3.0	2.8	3.1	2.9	(14)	
(15) Precipitation	PrecAvg	547	32	31	31	37	58	63	63	55	47	39	50	39	(15)	
	PrecMax	916	90	121	113	103	163	169	183	203	144	150	163	105	(16)	
	PrecMin	306	0	0	0	0	3	12	2	0	1	0	0	1	(17)	
	PrecStd	129	19	25	21	23	32	35	38	41	35	32	34	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	15.8	20.1	26.8	30.0	34.0	35.2	35.9	30.2	24.8	18.1	11.9	(19)	
		MCWB	8.9	10.6	12.2	16.6	20.3	22.7	21.7	22.2	19.5	18.0	13.6	8.7	(20)	
	2%	DB	9.1	12.9	17.8	23.2	28.0	31.9	33.0	33.1	27.9	22.2	15.2	9.1	(21)	
		MCWB	7.0	8.8	11.3	14.6	18.9	21.8	22.0	21.8	19.0	16.2	12.2	7.8	(22)	
	5%	DB	7.1	10.2	15.1	21.2	26.1	29.7	31.1	31.0	25.8	20.1	13.6	7.8	(23)	
		MCWB	5.7	7.4	9.8	13.6	18.0	20.9	20.8	21.2	17.9	15.2	11.1	6.7	(24)	
	10%	DB	5.8	8.2	13.1	19.2	24.0	27.8	29.1	28.9	23.2	18.0	11.9	6.1	(25)	
		MCWB	4.5	6.2	8.7	12.7	17.0	19.7	20.2	20.3	16.7	13.9	9.8	5.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	10.9	13.4	17.6	21.5	24.4	24.5	25.0	20.8	18.5	14.6	9.6	(27)	
		MCDB	11.6	14.7	18.2	25.0	28.5	31.9	32.1	33.1	27.6	23.3	17.1	10.5	(28)	
	2%	WB	7.5	9.3	11.8	15.6	19.7	22.6	23.0	23.1	19.6	17.0	12.9	7.9	(29)	
		MCDB	8.7	12.0	16.2	21.8	26.2	29.8	30.5	30.3	26.2	21.2	15.3	9.0	(30)	
	5%	WB	5.9	7.9	10.4	14.4	18.6	21.3	22.1	22.0	18.6	15.7	11.5	6.5	(31)	
		MCDB	6.8	9.9	14.0	20.0	24.6	27.7	28.7	28.7	24.1	19.2	13.3	7.2	(32)	
	10%	WB	4.3	6.5	9.3	13.1	17.4	20.3	21.1	21.0	17.5	14.5	10.1	5.2	(33)	
		MCDB	5.2	8.1	12.8	18.1	22.9	26.2	27.2	27.5	22.1	17.4	11.3	5.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	8.1	10.1	11.7	11.7	11.8	12.0	12.2	11.0	10.4	7.4	5.7	(35)	
		MCDBR	8.6	11.3	14.4	15.5	15.4	14.7	15.1	15.4	14.5	13.3	9.9	7.2	(36)	
	5% WB	MCWBR	6.7	8.0	9.1	8.2	7.5	6.3	5.8	6.3	6.7	7.8	6.9	5.9	(37)	
		MCWBR	8.1	10.3	12.6	14.0	14.1	13.3	13.1	13.6	12.8	12.3	9.1	6.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.330	0.352	0.389	0.438	0.427	0.428	0.441	0.440	0.415	0.383	0.358	0.328	(40)	
		taud	2.365	2.317	2.246	2.125	2.216	2.246	2.212	2.223	2.260	2.328	2.352	2.373	(41)	
	Edn at Noon	Edn at Noon	739	800	823	813	839	839	821	803	786	751	693	691	(42)	
		Edh at Noon	72	93	116	144	137	134	136	130	114	92	74	65	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.04	1.85	3.10	4.57	5.52	5.99	5.87	5.26	3.70	2.37	1.25	0.79	(44)	
	RadStd	RadStd	0.12	0.27	0.46	0.44	0.57	0.45	0.42	0.51	0.43	0.26	0.12	0.10	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (50 neighbors)		+0.59	N/A	N/A	N/A	+0.53	+0.54	-120	-175	+97	+48	

Nomenclature: See separate page